

Session Objectives

1. Integrate and understand macronutrients, their building blocks, micronutrients, and their recommended intake ranges and diet composition.
2. Integrate and understand calorie requirements and energy balance.
3. Examine and discuss the importance of reading food labels.
4. Analyze the roles of macronutrients and micronutrients in human metabolism and homeostasis.

Nutrition and Health Are Closely Related

“About half of all American adults—117 million individuals—have one or more preventable chronic diseases, many of which are related to poor eating and physical activity patterns.”

The Guidelines

Follow a healthy dietary pattern at every life stage.



The Guidelines





Follow a healthy dietary pattern at every life stage.

- At every life stage—infancy, toddlerhood, childhood, adolescence, adulthood, pregnancy, lactation, and older adulthood—it is never too early or too late to eat healthfully.
- **For about the first 6 months of life**, exclusively feed infants human milk. Continue to feed infants human milk through at least the first year of life, and longer if desired. Feed infants iron-fortified infant formula during the first year of life when human milk is unavailable. Provide infants with supplemental vitamin D beginning soon after birth.
- **At about 6 months**, introduce infants to nutrient-dense complementary foods. Introduce infants to potentially allergenic foods along with other complementary foods. Encourage infants and toddlers to consume a variety of foods from all food groups. Include foods rich in iron and zinc, particularly for infants fed human milk.
- **From 12 months through older adulthood**, follow a healthy dietary pattern across the lifespan to meet nutrient needs, help achieve a healthy body weight, and reduce the risk of chronic disease.



Focus on meeting food group needs with nutrient-dense foods and beverages, and stay within calorie limits.

An underlying premise of the *Dietary Guidelines* is that nutritional needs should be met primarily from foods and beverages—specifically, nutrient-dense foods and beverages. Nutrient-dense foods provide vitamins, minerals, and other health-promoting components and have no or little added sugars, saturated fat, and sodium.

A healthy dietary pattern consists of nutrient-dense forms of foods and beverages across all food groups, in recommended amounts, and within calorie limits.

The core elements that make up a healthy dietary pattern include:

- **Vegetables of all types**—dark green; red and orange; beans, peas, and lentils; starchy; and other vegetables
- **Fruits**, especially whole fruit
- **Grains**, at least half of which are whole grain
- **Dairy**, including fat-free or low-fat milk, yogurt, and cheese, and/or lactose-free versions and fortified soy beverages and yogurt as alternatives
- **Protein foods**, including lean meats, poultry, and eggs; seafood; beans, peas, and lentils; and nuts, seeds, and soy products
- **Oils**, including vegetable oils and oils in food, such as seafood and nuts

Making Nutrient-Dense Choices: One Meal At a Time



Typical Burrito Bowl
Total Calories = 1,120

White rice (1 ½ cups)

Black beans (⅓ cup)

Chicken cooked with sauce (2 ounces)

No grilled vegetables

Guacamole (½ cup)

Jarred salsa (¼ cup)

Sour cream (¼ cup)

Cheese (⅓ cup)

Jalapeño (5 slices)

Iced tea with sugar (16 ounces)



Nutrient-Dense Burrito Bowl
Total Calories = 715

Brown rice (1 cup) + Romaine lettuce (½ cup)

Black beans, reduced sodium (⅓ cup)

Grilled chicken with spice rub (2 ounces)

Added grilled vegetables (⅓ cup)

Sliced avocado (5 slices)

Fresh salsa/pico de gallo (¼ cup)

No sour cream

Reduced-fat cheese (⅓ cup)

Jalapeño (5 slices)

Iced tea, no sugar (16 ounces)



Limit foods and beverages higher in added sugars, saturated fat, and sodium, and limit alcoholic beverages.

At every life stage, meeting food group recommendations—even with nutrient-dense choices—requires most of a person’s daily calorie needs and sodium limits. A healthy dietary pattern doesn’t have much room for extra added sugars, saturated fat, or sodium—or for alcoholic beverages. A small amount of added sugars, saturated fat, or sodium can be added to nutrient-dense foods and beverages to help meet food group recommendations, but foods and beverages high in these components should be limited.

Limits are:

- **Added sugars**—Less than 10 percent of calories per day starting at age 2. Avoid foods and beverages with added sugars for those younger than age 2.
- **Saturated fat**—Less than 10 percent of calories per day starting at age 2.
- **Sodium**—Less than 2,300 milligrams per day—and even less for children younger than age 14.
- **Alcoholic beverages**—Adults of legal drinking age can choose not to drink or to drink in moderation by limiting intake to 2 drinks or less in a day for men and 1 drink or less in a day for women, when alcohol is consumed. Drinking less is better for health than drinking more. There are some adults who should not drink alcohol, such as women who are pregnant.

The *Dietary Guidelines for Americans, 2020-2025* includes nearly all of the science-based recommendations of the 2020 Dietary Guidelines Advisory Committee

- The 2020-2025 edition of the *Dietary Guidelines* emphasizes the importance of limiting intakes of added sugars and alcoholic beverages as recommended by the Committee, but does not include changes to quantitative recommendations.
- Any revisions to previous editions of the *Dietary Guidelines* must have sufficient scientific justification, and by law, must be based on the preponderance of scientific and medical knowledge current at the time.
- The preponderance of evidence supports limiting intakes of added sugars and alcoholic beverages to promote health and prevent disease; however, the evidence reviewed since the 2015-2020 edition does not substantiate quantitative changes at this time.
- USDA and HHS encourage more research on the relationship between added sugars and alcoholic beverages and health, and plan to monitor these topics.

Chapter 2. Infants and Toddlers

Key Recommendations

- **At about 6 months**, introduce infants to nutrient-dense complementary foods.
- Introduce infants to potentially allergenic foods along with other complementary foods.
- Encourage infants and toddlers to consume a variety of foods from all food groups. Include foods rich in iron and zinc, particularly for infants fed human milk.
- Avoid foods and beverages with added sugars.
- Limit foods and beverages higher in sodium.
- As infants wean from human milk or infant formula, transition to a healthy dietary pattern.

**For lifelong good health, make every bite count
with the *Dietary Guidelines for Americans***



The *Dietary Guidelines for Americans*:

What It Is, What It Is Not

- Provide ***evidence-based recommendations*** about the components of a healthy and nutritionally adequate diet
- ***Focus on disease prevention*** rather than disease *treatment*
- Inform Federal food, nutrition, and health policies and programs

Key Elements of Healthy Eating Patterns:

Key Recommendations

 **Consume a healthy eating pattern that accounts for all foods and beverages *within an appropriate calorie level*.**

 **A healthy eating pattern *includes*:**

- A variety of vegetables from all of the subgroups—dark green, red and orange, legumes (beans and peas), starchy, and other
- Fruits, especially whole fruits
- Grains, at least half of which are whole grains
- Fat-free or low-fat dairy, including milk, yogurt, cheese, and/or fortified soy beverages
- A variety of protein foods, including seafood, lean meats and poultry, eggs, legumes (beans and peas), and nuts, seeds, and soy products
- Oils

 **A healthy eating pattern *limits*:**

- Saturated fats and *trans* fats, added sugars, and sodium

Key Elements of Healthy Eating Patterns:

Key Recommendations (cont.)

Key Recommendations that are quantitative are provided for several components of the diet that should be limited. These components are of particular public health concern in the United States, and the specified limits can help individuals achieve healthy eating patterns within calorie limits:

- ❦ ***Consume less than 10 percent of calories per day from added sugars***
- ❦ ***Consume less than 10 percent of calories per day from saturated fats***
- ❦ ***Consume less than 2,300 milligrams (mg) per day of sodium***
- ❦ If alcohol is consumed, it should be consumed in moderation—up to one drink per day for women and up to two drinks per day for men—and only by adults of legal drinking age
- ❦ Meet the *Physical Activity Guidelines for Americans*

Inside Healthy Eating Patterns: Food Groups

“Eating an appropriate mix of foods from the food groups and subgroups—within an appropriate calorie level—is important to promote health.”

- Each food group and subgroup provides an array of nutrients.
- Recommended amounts reflect eating patterns associated with positive health outcomes.
- Foods from all of the food groups should be eaten in nutrient-dense forms.

Inside Healthy Eating Patterns:

Food Groups — Examples of Content



Vegetables

- Healthy eating patterns include a variety of vegetables from all five vegetable subgroups— **dark green**, **red and orange**, legumes (beans and peas), starchy, and other.
- Include all fresh, frozen, canned, and dried options in cooked or raw forms, including vegetable juices.
- Vegetables should be consumed in a nutrient-dense form, with ***limited additions such as salt, butter, or creamy sauces.***



Dairy

- Healthy eating patterns include fat-free and low-fat (1%) dairy, including milk, yogurt, cheese, or fortified soy beverages (soymilk).
- Fat-free or low-fat milk and yogurt, in comparison to cheese, contain less saturated fats and sodium and more potassium, vitamin A, and vitamin D.

Inside Healthy Eating Patterns:

Other Components

“In addition to the food groups, it is important to consider other food components when making food and beverage choices.”

- Added sugars
- Saturated fats
- *Trans* fats
- Cholesterol
- Sodium
- Alcohol
- Caffeine

Inside Healthy Eating Patterns:

Other Components — Examples of Content



Cholesterol

- Individuals should eat as little dietary cholesterol as possible while consuming a healthy eating pattern.
- The Healthy U.S.-Style Eating Pattern contains ***approximately 100 to 300 mg of cholesterol*** across the 12 calorie levels.



Caffeine

- Much of the available evidence on caffeine focuses on coffee intake.
- ***Three to five 8-oz cups of coffee per day*** can be incorporated into healthy eating patterns.
- Individuals who do not consume caffeinated beverages are not encouraged to incorporate them into their eating pattern.
- In addition, caffeinated beverages may contain added calories from ***cream, whole or 2% milk, creamer, and added sugars, which should be limited.***

Shifts to Align with Healthy Eating Patterns: Examples

- **Increasing vegetables** in mixed dishes while **decreasing the amounts of refined grains or meats high in saturated fat and/or sodium**.
- **Incorporating seafood** in meals twice per week in place of meat, poultry, or eggs.
- **Using vegetable oil** in place of solid fats when cooking, and using oil-based dressings and spreads on foods instead of those made from solid fats.
- Choosing beverages with **no added sugars**, such as water.
- Using the Nutrition Facts label to compare **sodium** content of foods and choosing the product with less sodium.

Empty Calories

Foods with lots of calories from fats and sugars and little or no nutrients.



Social Environment

- Eating on the run
- Eating as recreation
- Pressure to be sedentary
- Fewer meals at home
- Powerful and constant advertising
- Society of spectators instead of participant

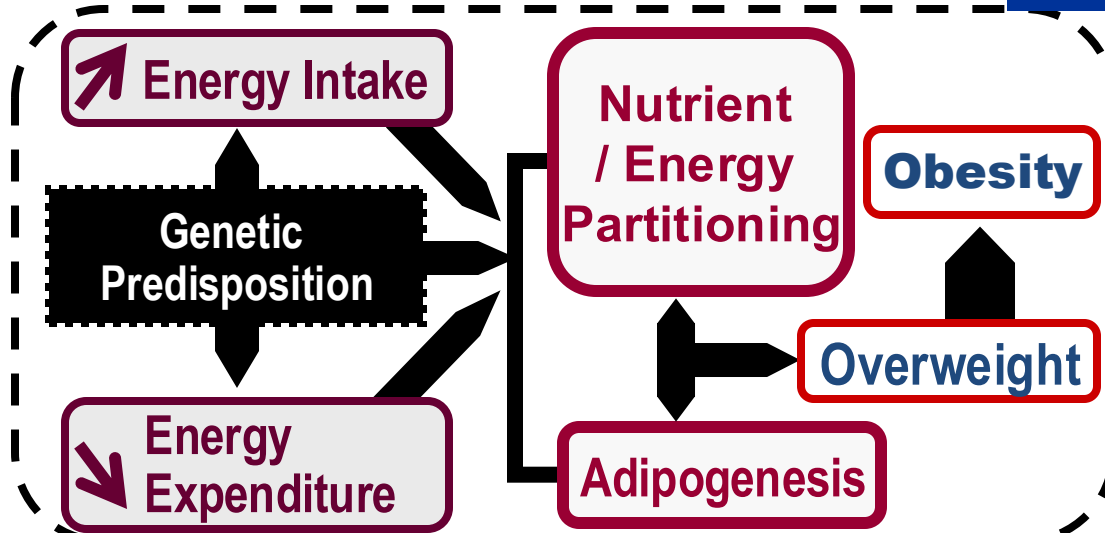
Physical Environment

- Pressure to consume
- Building design
- Absence of sidewalks
- Population density
- Urban sprawl
- Pollutants
- Automobile dependence
- Certain medications

- Calorie-dense foods
- Maternal-fetal nutrition
- More sedentarism
- Lactation
- High fat diets
- Larger portions

Behavior

- Smoking cessation
- Less physical activity
- Corn fructose syrup



Thermogenesis

n-6/n-3 PUFAs

Lipid ox

Viruses

RMR

Regulators of adipogenesis: RAR, RXR, PPAR γ , C/EBP, SREBP-1c, PGC-1, etc.

Genetic hypotheses

Epigenetics

Biology

CNS regulators of appetite: NPY, α -MSH, CART, Orexins, Agouti, MC4R, MCH, AGRP, etc.

Peripheral regulators of appetite: PYY, insulin, leptin, ghrelin, CCK, GLP-1, etc.

Macronutrients – Energy Producing

- **Carbohydrates (CHO)**

- Provides energy

- **Proteins**

- Builds/repairs body tissue

- **Fats (lipids)**

- Provides insulation/energy, protects organs

- Important for growth and development
- Act to keep body functioning normally
- Provide energy (*kilocalorie*)

- Kcal: measure of the amount of heat needed to raise the temperature of 1000 grams (1 liter) of water to 1 degree C



Caloric Breakdown

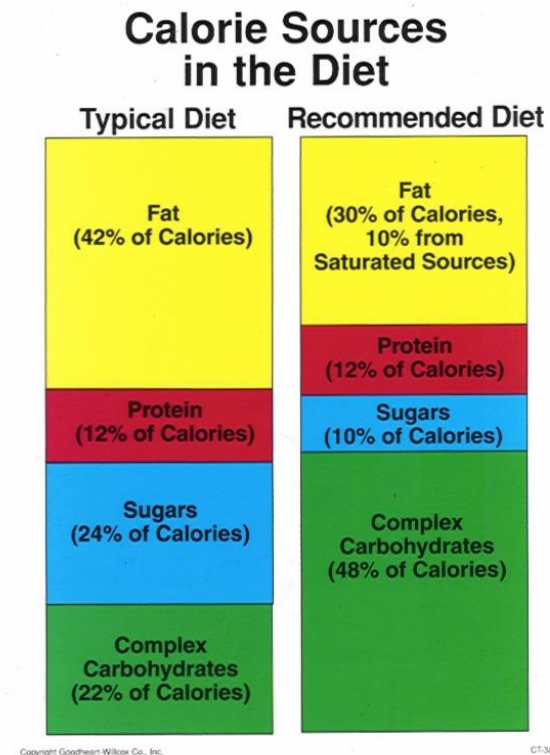
Carbohydrates: 55-60%

Fat: No more than 30%

Protein: 10-15 %

The Average American:

- Consumes too many calories from saturated fat, sugar, & sodium.
- Doesn't consume enough fiber.



Carbohydrates - CHO (Starch, Sugar, Fiber)

SIMPLE CHO

- **Monosaccharides** (one unit sugars)
 - Glucose, fructose, galactose
 - Absorbed directly, no breakdown
- **Disaccharides** (formed when 2 monosaccharides combine)
 - Lactose: glucose and galactose (found in milk and dairy)
 - Sucrose: glucose and fructose (table sugar, also naturally in some fruits and other foods)



COMPLEX CHO

- **Polysaccharides** - many units of sugar
 - Both Fiber and Starch
 - *Plant foods, grains, potatoes, and legumes contain starches*
 - All CHO except for fiber are broken down to simple sugars
 - **Fiber:**
 - *Not entirely broken down*
 - *Some stays in digestive system*
 - *Health benefits*



2. Focus on variety, nutrient density, and amount.

- o To meet nutrient needs within calorie limits, *choose a variety of nutrient-dense foods* across and within all food groups in recommended amounts.
- o Nutrient dense foods provide vitamins, minerals and other beneficial substances with relatively few calories.

Which is more Nutrient Dense?

Spinach



OR



Candy

Summary Slide

- **Saturated fat:** <10% of diet [beginning at age 2]
- **Sodium:** < 2300 mg/day [Ages 14 and older]
 - Or even less if younger than age 14]
- **Sugar:** <100 cal/day female [<150 male] → <10% diet [Ages 2 and older]
 - AVOID for infants and toddlers
- **Alcohol [if consumed]:** Limiting to 2 drinks or less a day for men and 1 drink or less a day for women.
- **BMI, Waist Circumference and Disease Risk** → See Table

Added Sugars: Intakes and Limit

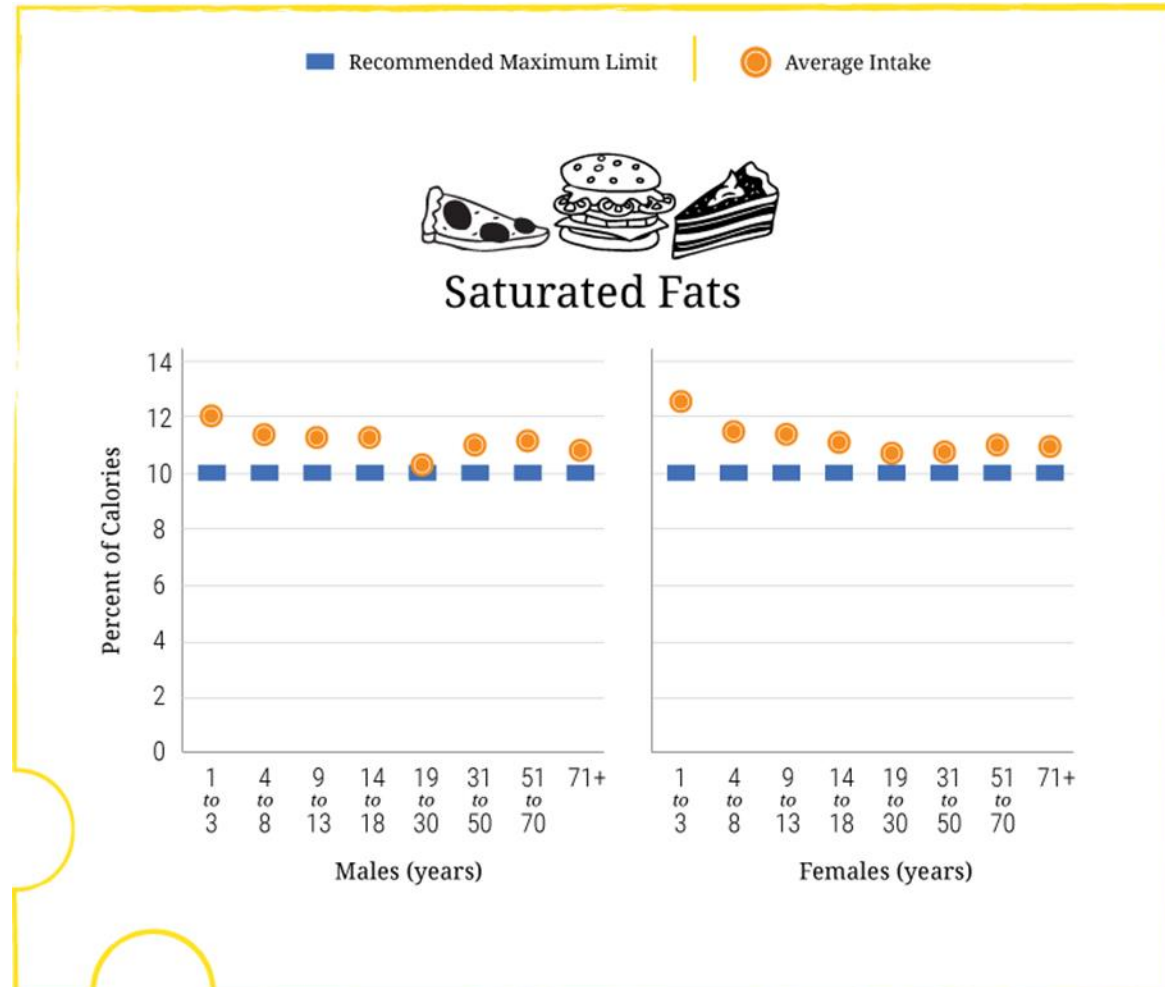
Average Intakes as a Percent of Calories per Day by Age-Sex Group, in Comparison to the Dietary Guidelines Maximum Limit of Less than 10 Percent of Calories (Figure 2-9)



- **Note:**
The maximum amount of added sugars allowable in a Healthy U.S.-

Saturated Fats: Intakes and Limit

Average Intakes as a Percent of Calories per Day by Age-Sex Group, in Comparison to the Dietary Guidelines Maximum Limit of Less Than 10 Percent of Calories (Figure 2-11)



- **Data Source:** What We Eat in America, NHANES 2007-2010 for average